Body Fluids and Circulation

Blood and Lymph

1. Erythrocytes, leucocytes and platelets are collectively called formed elements and they constitute nearly _____ percent of blood. Fill in the blank.
1) 45
2) 75
3) 25
4) 90

2. Identify the wrong statement w.r.t. leucocytes.
1) They are colourless due to lack of haemoglobin.
2) They are devoid of nucleus.
3) Lymphocytes and monocytes are agranulocytes.
4) Neutrophils, basophils and eosinophils are different types of granulocytes.

3. Plasma without the clotting factors is called
1) Lymph
2) Blood
3) Serum
4) Interstitial fluid

4. Which of the following is an incorrect statement w.r.t. blood plasma?
1) It is a straw coloured, viscous fluid constituting 55% of blood.
2) Fibrinogen, globulins and albumins are the major plasma proteins.
3) Plasma also has small amount of minerals like Na^+, Ca^{2+}, Mg^{2+}, HCO_3 Cl^- etc.
4) Factor for coagulation or clotting of blood are also present in the plasma in active form.

5. _____ are most abundant WBCs in blood, while ____ are the least abundant WBCs. Fill in the blanks.
1) Neutrophils, lymphocytes
2) Neutrophils, basophils
3) Monocytes, lymphocytes
4) Lymphocytes, eosinophils

6. RBCs have an average life span of 120 days-after which they are destroyed in _____ . Fill in the blanks.
1) Spleen
2) Pancreas
3) Kidney
4) Stomach

7. Identify the wrong statement w.r.t. human RBCs.
1) They are devoid of nucleus.
2) They are biconvex in shape.
3) They have iron-containing complex protein called haemoglobin.
4) They are formed in the red bone marrow in adults.

8. _____ and _____ are phagocytic cells which destroy foreign organisms entering the body: Fill in the blanks.

1) Neutrophils, lymphocytes
2) Neutrophils, monocytes
3) Basophils, lymphocytes
4) Eosinophils, lymphocytes

9. Match the options given in column I with those in column II.

	Column I		Column II
A	Albumin	(i)	Defence mechanism
B	Globulin	(ii)	Osmotic balance
C	Fibrinogen	(iii)	Blood coagulation

1) A-(i), B-(ii), C-(iii)
2) A-(ii), B-(i), C-(iii)
3) A-(iii), B-(i), C-(ii)
4) A-(iii), B-(ii), C-(i)

10. Which of the following option is correct for interstitial fluid?

1) Water + Small water soluble substances
2) Blood - Larger proteins
3) Blood - Formed elements
4) Water + RBCs + Small water soluble substances

11. Read the following statements.

(a) Plasma has 90-92% water and 6-8% proteins.
(b) A healthy adult man has, on an average, 5-5.5 million of RBCs mm^{-3} of blood.
(c) Granulocytes include basophils, neutrophils and monocytes.
(d) Reduction in number of platelets may cause excessive blood loss.
(e) Erythrocytes are the most abundant of all the cells in our blood.

Choose the option with incorrect statements.

1) a and b
2) c only
3) c and d
4) b, d and e

12. ____ resist infections and are also associated with allergic reactions. Fill in the blank.

1) Lymphocytes
2) Neutrophils
3) Eosinophils
4) Thrombocytes

13. The normal level of haemoglobin in blood is _____ gms in every 100 mL of blood. Fill in the blank.

1) 30-40
2) 12-16
3) 4-6
4) 20-25

14. Which of the following statements are incorrect?

(i) Important role of lymphocytes is to produce antibodies.
(ii) Neutrophils have multilobed nucleus.
(iii) Basophils are formed by breakdown of megakaryocytes.

(iv) Monocytes have large round nucleus.
(v) The normal WBC count, in blood is 6000-8000 mm^{-3}
1) (i) and (ii)
2) (ii) and (iii)
3) (iii) and (iv)
4) (iv) and (v)

15. Read the following statements:
(i) RBCs plays a significant role in transport of respiratory gases.
(ii) Lymph has abundant WBCs and mostly lymphocytes.
(iii) Blood normally contains 1,50,000 3,50,000 platelets mm^{-3}.
(iv) Lymphocytes are special cells in the bone marrow.
(v) Serum is plasma with clotting factors like fibrinogen.
Which of the above statements are incorrect?
1) (i) and (ii)
2) (ii) and (iii)
3) (iii) and (iv)
4) (iv) and (v)

16. Which of the following are the largest WBCs?
1) Monocytes
2) Neutrophils
3) Basophils
4) Lymphocytes

17. The blood corpuscles lacking nuclei are
1) Platelets and neutrophils
2) Platelets and erythrocytes
3) Erythrocytes and lymphocytes
4) Basophils and eosinophils

18. Phagocytic activity is not exhibited by
1) Neutrophils
2) Monocytes
3) Macrophages
4) Basophils

19. ____ secrete histamine, serotonin, heparin, etc. and are involved in inflammatory reactions. Fill in the blank.
1) Eosinophils
2) Basophils
3) Neutrophils
4) Monocytes

20. Regarding erythroblastosis foetalis, which of the following statement is correct?
1) This could be fatal to the mother only.
2) Could cause severe anaemia and jaundice to the mother.
3) This is also called haemolytic disease of newborn.
4) All of the above

21. The role of enzyme thrombokinase is
1) Conversion of fibrinogen into fibrin
2) Conversion of prothrombin into thrombin

3) Formation of urea in liver
4) Cellular respiration

22. Identify the wrong statement w.r.t. blood groups.
1) Rhesus antigen is present on the surface of RBCs.
2) Blood group AB is the universal recipient.
3) Person with blood group O do not have antibodies in their plasma.
4) Persons with blood group AB do not have antibodies in their plasma.

Circulatory Pathways

23. Mitral valve is present between
1) Left atrium and right ventricle
2) Left atrium and left ventricle
3) Left ventricle and right ventricle
4) Right atrium and right ventricle

24. 'Bundle of His' in heart refers to
1) Nervous tissue supplying ventricles
2) Muscular tissue supplying ventricles
3) Muscular tissue supplying atria
4) Nerves regulating opening and closing of atrioventricular valves

25. Stethoscope is used to hear heart sounds produced during cardiac cycle. The second heart sound is heard when
1) AV valves open
2) AV node receives signal from SA node
3) Atria contract to push blood into ventricles
4) Semilunar valves close down after the blood flows into vessels from ventricles

26. Which of the following is a correct statement?
1) Stroke volume divided by heart rate gives the cardiac output.
2) Cardiac output divided by heart rate gives the stroke volume.
3) Heart rate multiplied by cardiac output gives the stroke volume.
4) Stroke volume multiplied by cardiac output gives the heart rate.

27. If due to some injury the chordae tendineae of the tricuspid valve of the human heart becomes nonfunctional partially, then it will result in
1) Slowing down of blood flow into aorta
2) Slowing down of blood flow into pulmonary artery
3) Backflow of blood into left atrium
4) Slowing down of impulse generation by SA node

28. The first heart sound is produced
1) When semilunar valves close
2) In the beginning of atrial systole
3) In the beginning of ventricular systole due to closing down of AV valves
4) In the beginning of ventricular diastole

29. Out of curiosity, Aditya got his regular medical tests done along with his ECG. The depolarisation and repolarisation process are represented during the cardiac cycle. Aditya has no cardiac issues thus his ECG was normal. Which one the following waves must not be present in his graph?

1) Depolarisation of ventricles
2) Depolarisation of atria
3) Repolarisation of atria
4) Repolarisation of ventricles

30. The volume of blood pumped by each ventricle during a cardiac cycle is
1) Called stroke volume
2) Around 70 mL
3) Reduced in patients with heart failure
4) All of the above

31. Select the incorrect statements from the following:
1) Molluscs and arthropods have open circulatory system.
2) Single circulation is seen in Pisces.
3) Birds and mammals have incomplete double circulation.
4) Birds and mammals have incomplete double circulation.

32. Identify the wrong statements w.r.t. human circulatory system.
1) In a healthy man, the cardiac output is 5000 mL.
2) Chordae tendineae are fibrous cords that join the flaps of atrioventricular valves with papillary muscles of ventricles.
3) The opening of superior vena cava into the right atrium is guarded by mitral valve.
4) Semilunar valves are located at the base of pulmonary artery and aorta.

33. Select the incorrect statement from the following:
1) The wall of ventricles is thicker than the wall of atria.
2) The two prominent heart sounds can be easily heard through a stethoscope.
3) Contraction of right ventricle pump blood into the systemic aorta.
4) All vertebrates have muscular chambered heart.

34. Which of the following is an incorrect statement w.r.t. the events of cardiac cycle?
1) When all the four chambers of heart are in a relaxed state, they are in joint diastole.
2) Fall in ventricular pressure leads to closing of semilunar valves in the beginning of ventricular diastole
3) The walls of ventricles are much thicker than that of the atria.
4) The atrioventricular valves remain open during ventricular systole.

35. During every cardiac cycle, which statement stands true?
1) Aorta and pulmonary artery receive different volume of blood.
2) Left and right ventricle pump same volume of blood.
3) Left and right ventricle pump different volumes of blood
4) Each atrium receives different volume of blood.

36. Read the following statements and choose the correct option:

Statement I: The simultaneous contraction of both the atria causes atrial systole which increases the flow of blood into ventricles by about 30 percent.

Statement II: During ventricular systole, oxygenated blood is pumped into aorta while deoxygenated blood is pumped into pulmonary arteries.
1) Both statements I and II are correct
2) Both statements I and II are incorrect
3) Only statement I is correct
4) Only statement II is correct

37. Which of the following is an incorrect statement w.r.t. SA node?

1) It generates impulse with greatest rhythmicity.
2) It is called the pacemaker of heart.
3) It is a mass of neural tissue.
4) It is located in right atrium.

38. During pulmonary circulation
1) Oxygenated blood is carried by pulmonary veins to left atrium
2) Deoxygenated blood is pumped by right ventricle to lungs through pulmonary arteries
3) Deoxygenated blood returns to right atrium through vena cavae
4) Both (1) and (2)

39. Find the mismatch from the following:
1) Left atrium-Receives oxygenated blood from lungs through two pulmonary veins
2) Right ventricle-Pumps deoxygenated blood to lungs through pulmonary arteries
3) Right atrium-Receives deoxygenated blood from coronary sinus, and the two vena cavae
4) Left ventricle-Pumps oxygenated blood to the body through aorta

40. The part of conducting system present in ventricular wall of human heart is
1) Sinoatrial node
2) Purkinje fibres
3) Atrioventricular node
4) Internodal pathways

41. The closing of semilunar valve does not coincide with
1) Fall in ventricular pressure
2) Production of dub sound
3) Ventricular diastole
4) Ventricular diastole

42. Which of the following is the correct pathway for the conduction of impulse in heart?
1) SA node-AV bundle-AV node-Purkinje fibres
2) AV node—SA node-AV bundle-Purkinje fibres
3) SA node-AV node-bundle of His-Purkinje fibres
4) AV bundle-SA node-AV node-Purkinje fibres

43. Blood from head, neck and arms enter the heart through
1) Superior vena cava
2) Inferior vena cava
3) Pulmonary vein
4) Pulmonary artery

44. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): By counting the number of QRS complexes that occur in a given time period, one can determine the heart beat rate of an individual.

Reason (R): The ECGs obtained from different individuals have roughly the same shape for a given lead configuration.

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is incorrect.
3) (A) is incorrect but (R) correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

45. Which of the following is incorrect w.rt. properties of veins?
1) They have narrow lumen.
2) They have valve to check backflow of blood.
3) They mostly carry deoxygenated blood.
4) All of the above.

46. Given below are two statements:
Statement I: Endothelium lining of blood vessels has simple squamous epithelium.
Statement II: Capillaries are the smallest blood vessels in human body.
In the light of the above statements, choose the most appropriate answer from the options given below:
1) Both statement I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, and statement II is correct.
4) Both statement I and statement II are correct.

47. The wall of artery is _____ and blood moves at _____ pressure. Fill in the blanks.
1) Thick, low
2) Thin, high
3) Thick, high
4) Thin, low

48. The assumption that the cardiac activity could be moderated by the autonomous neural system is supported because
1) The heart rate is decreased but the stroke volume is increased by the sympathetic system.
2) The heart rate and stroke volume are stimulated by the sympathetic system.
3) The heart rate and stroke volume are stimulated by the parasympathetic system.
4) The heart rate is decreased but the stroke volume is increaşed by the parasympathetic system.

49. Which one of the following blood vessel carries blood from intestine to the liver before it is delivered to systemic circulation?
1) Hepatic portal vein
2) Hepatic vein
3) Subclavian vein
4) Jugular vein

50. Read the following statements:
(i) Pulse can be felt on a vein.
(ii) The inner lining of blood vessel is called endothelium.
(iii) The cardiac tissue present in the wall of heart are supplied by coronary artery.
(iv) Both renal and hepatic portal systems are present in humans.
Which of the above statements are correct?
1) (i) and (ii)
2) (ii) and (iii)
3) (iii) and (iv)
4) (i) and (iv)

51. Given below are two statements:
Statement I: Sympathetic nervous system can increase the strength of ventricular contraction and cardiac output.

Statement II: Adrenal medullary hormones can decrease the cardiac output.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both statement I and II are incorrect.
2) Statement I is correct, but statement II is incorrect.
3) Statement I is incorrect, and statement II is correct.
4) Both statement I and statement II are correct.

52. Match the following columns and choose the correct option.

	Column I		Column II
1	Tunica intima	a	Smooth muscle
2	Tunica media	b	Fibrous connective tissue
3	Tunica externa	c	Squamous endothelium
4	Cardiac arrest	d	Heart is not pumping blood effectively
5	Cardiac arrest	e	Hear stops beating

1) 1 - c, 2 - b, 3- a, 4-e, 5-d
2) 1-a, 2-c, 3-b, 4-d, 5-e
3) 1-c, 2- a, 3-b, 4- e, 5- d
4) 1-b, 2-a, 3-c, 4-d, 5-e

53. Normal activities of the heart are

1) Regulated intrinsically
2) Auto regulated by specialised muscles
3) Regulated by nodal tissues
4) All of the above

54. A special neural centre in the medulla oblongata can moderate the cardiac function through

1) Autonomic nervous system
2) Somatic neural centre
3) Sympathetic and parasympathetic neural system
4) Both (1) and (3)

55. Read the following statements:

Statement I: A portal vein does not carry blood directly to heart but forms a network of capillaries in another or intermediate organ before reaching the heart.

Statement II: Hepatic and hypophysial portal systems are present in humans

Select the correct option:

1) Both statements are correct
2) Both statements are incorrect
3) Only statements I is correct
4) Only statement II is correct

56. Read the following statement.

i. The systemic circulation provides nutrients, O_2 and other essential substances to the tissues and takes CO_2 and other harmful substances away for elimination.
ii. The oxygenated blood entering the aorta is carried by a network of arteries, arterioles and capillaries to the tissues.
iii. Angina can occur in men and women of any age, but it is more common among the middle-aged persons and elderly.
iv. Angina can occur in men and women of any age, but it is more common among the middle-aged persons and elderly.

How many statements are correct?

1) Two
2) Three
3) Four
4) One

57. The deoxygenated blood pumped into the pulmonary artery is passed on to the lungs from where the oxygenated blood is carried by the pulmonary veins into the left atrium. This pathway constitutes the

1) Systemic circulation
2) Pulmonary circulation
3) Coronary circulation
4) Hepatic portal circulation

58. Human heart is termed myogenic because

1) Sympathetic nerves can increase the heart rate
2) Heart has thick layer of cardiac muscle in its wall
3) Impulse for heart beat is generated in the pacemaker
4) Impulse for heart beat is generated outside the heart

59. Select the correct statement

1) Right ventriele receives oxygenated blood through vena cava.
2) Left atrium receives oxygenated blood from pulmonary artery.
3) Right atrium receives deoxygenated blood from aorta.
4) None of the above

60. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Coronary Artery Disease (CAD), also called angina affects blood vessels.

Reason (R): CAD is caused due to Ca^{++}, fat, cholesterol and fibrous tissue deposition in wall of artery.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
2) (A) is correct but (R) is incorrect.
3) (A) is incorrect but (R) correct.
4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

61. Artificial pacemaker is used for

1) Heart attack
2) Atherosclerosis
3) Angina pectoris
4) Irregularity in heart rhythm

62. Which of the following is an incorrect statement?

1) Rise in blood pressure beyond 140 mmHg (systolic) and 90 mmHg (diastolic) is called hypertension.
2) Congestion of lungs is seen in patients with heart failure.
3) Interrupted blood flow to the walls of heart causes stroke.
4) The term hypertension is related to blood pressure higher than 120/80 mmHg.

63. Patients with angina pectoris have acute chest pain when

1) They have high blood sugar

2) They have low blood pressure
3) Enough oxygen does not reach heart muscles
4) Heart muscles contract rapidly in an uncoordinated manner.

64. In a healthy individual, the average diastolic and systolic pressure is
1) 120 mm Hg and 100 mm Hg respectively
2) 80 mm Hg and 120 mm Hg respectively
3) 120 mm Hg and 80 mm Hg respectively
4) 100 mm Hg and 120 mm Hg respectively

65. Which among the following statement is incorrect?
1) Angina pectoris is characterized by acute chest pain due to injury to the ribcage.
2) Tunica media of blood vessels is rich in smooth muscles.
3) High blood pressure leads to heart diseases and also affects vital organs like brain and kidney.
4) A special coronary system of blood vessels is present in our body exclusively for the circulation of blood to and from the cardiac musculature.

66. Which of the following is disorder of circulatory system?
1) Rhinitis
2) Atherosclerosis
3) Emphysema
4) Osteoporosis

67. Select the incorrect statement.
1) If repeated checks of blood pressure of an individual is 140/90 mmHg, it shows hypertension.
2) Angina pectoris is common in middle aged and elderly people.
3) Heart failure is same as cardiac arrest.
4) Atherosclerosis affects the vessels that supply to heart muscles.

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
1	2	3	4	2	1	2	2	2	1	2	3	2	3	4
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
1	2	4	2	3	2	3	2	2	4	2	2	3	3	4
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
3	3	3	4	2	1	3	4	1	2	4	3	1	4	1
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
4	3	2	1	2	2	3	4	4	1	3	2	3	4	3
61	**62**	**63**	**64**	**65**	**66**	**67**								
4	3	3	2	1	2	3								